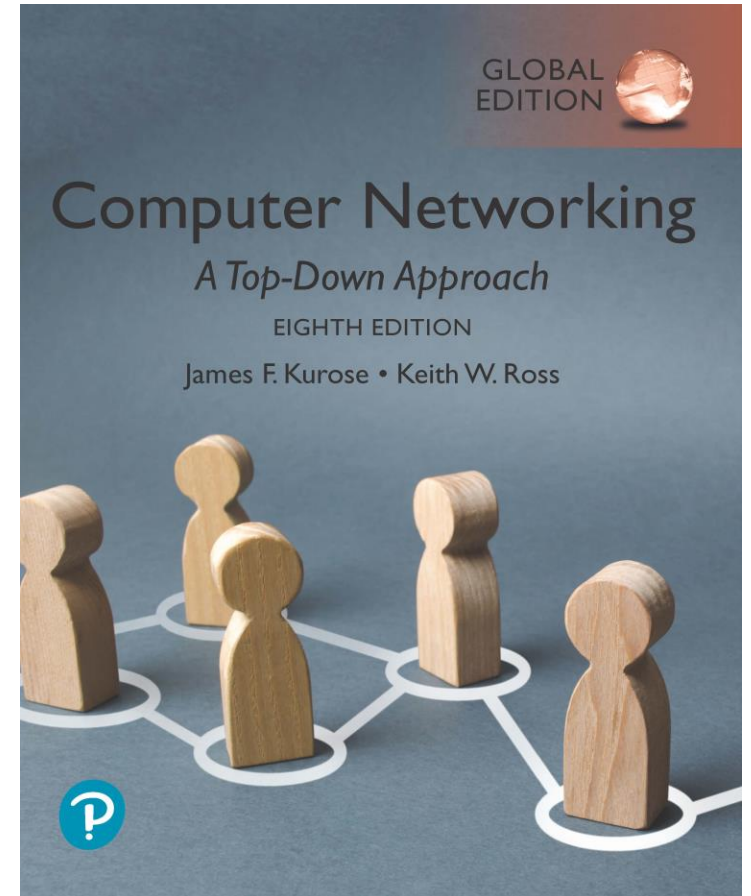


Chapter 3

Transport Layer



Computer Networking: A Top-Down Approach

8th Edition, Global Edition

Jim Kurose, Keith Ross

Copyright © 2022 Pearson Education Ltd

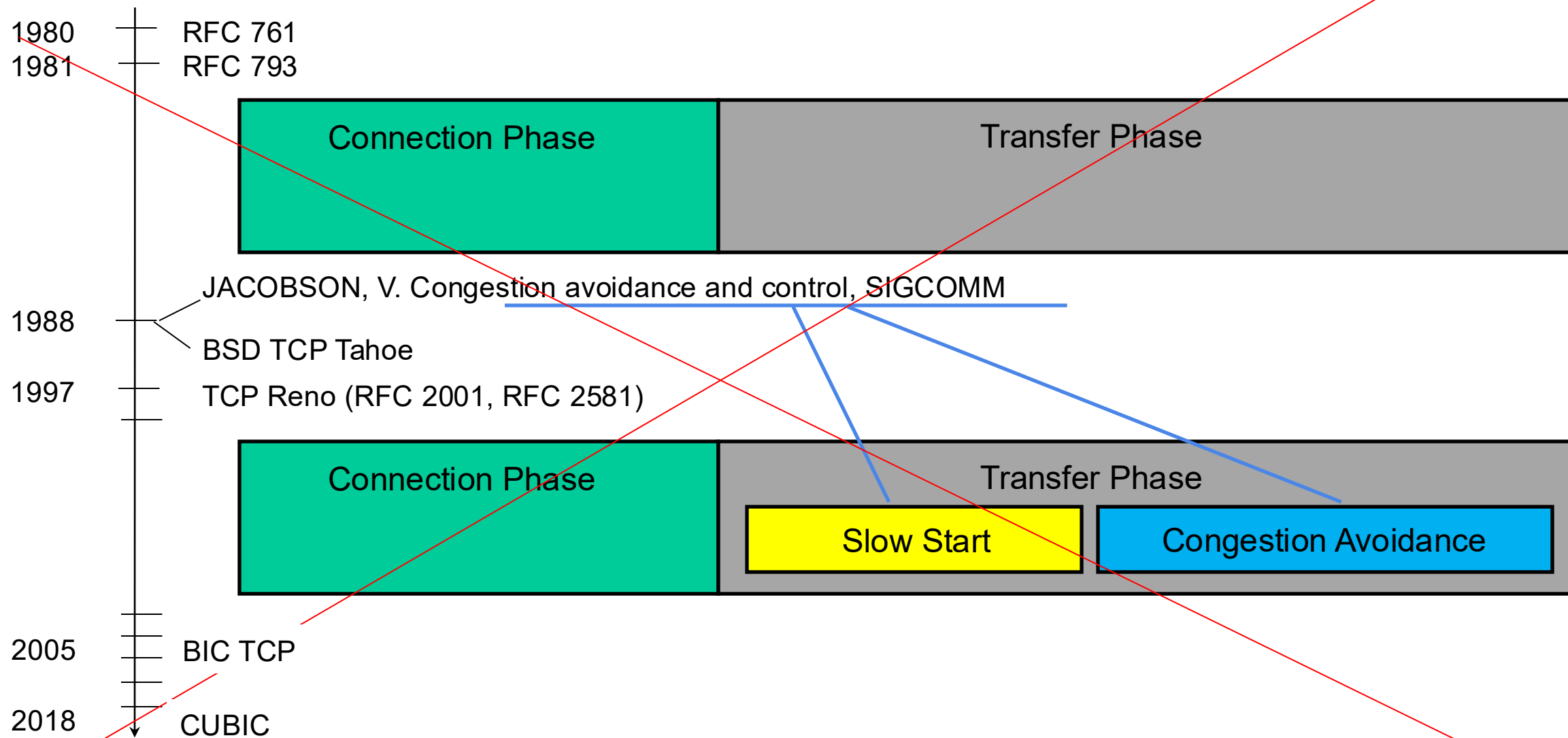
Transport layer: roadmap

- Transport-layer services
- Multiplexing and demultiplexing
- Connectionless transport: UDP
- Principles of reliable data transfer
- Connection-oriented transport: TCP
- Principles of congestion control
- TCP congestion control
- Evolution of transport-layer functionality



The major changes of TCP

Not part of course



Evolving transport-layer functionality

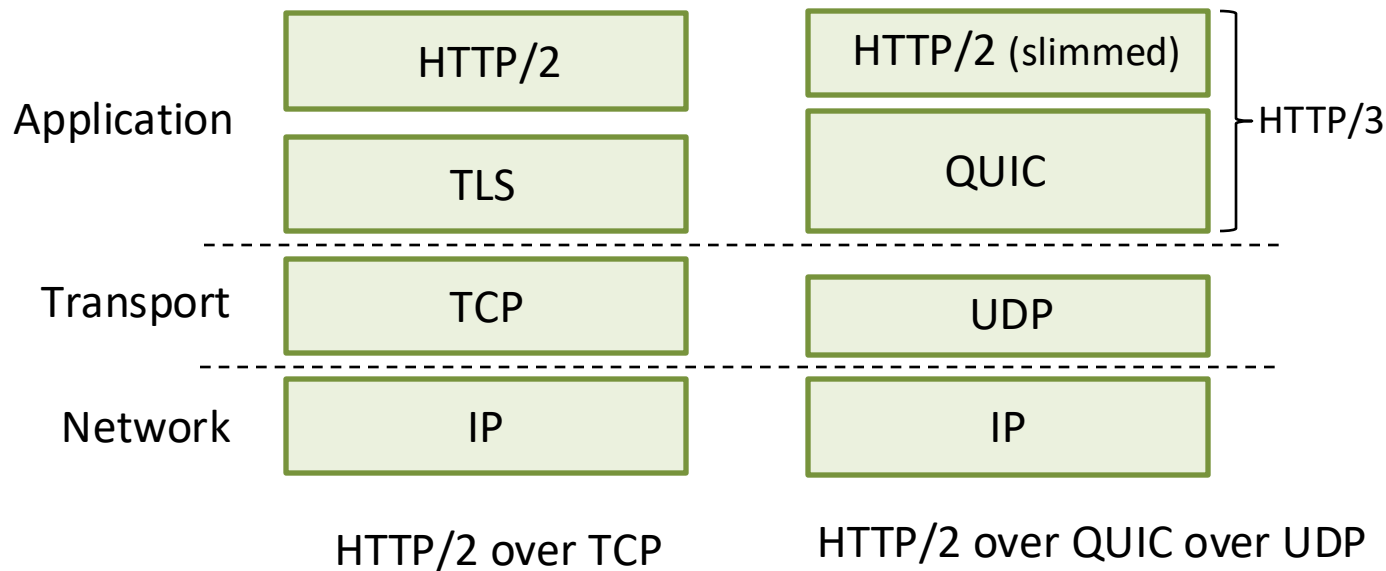
- TCP, UDP: principal transport protocols for 40 years
- different “flavors” of TCP developed, for specific scenarios:

Scenario	Challenges
Long, fat pipes (large data transfers)	Many packets “in flight”; loss shuts down pipeline
Wireless networks	Loss due to noisy wireless links, mobility; TCP treat this as congestion loss
Long-delay links	Extremely long RTTs
Data center networks	Latency sensitive
Background traffic flows	Low priority, “background” TCP flows

- moving transport–layer functions to application layer, on top of UDP
 - HTTP/3: QUIC

QUIC: Quick UDP Internet Connections

- application-layer protocol, on top of UDP
 - increase performance of HTTP
 - deployed on many Google servers, apps (Chrome, mobile YouTube app)

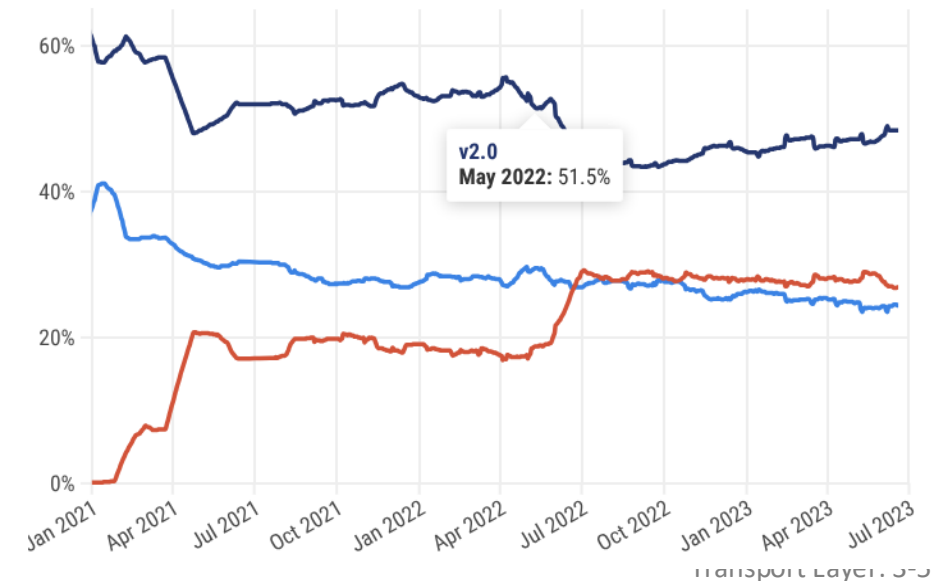


H/3 Adoption Grows Rapidly

HTTP Versions In Use 2021 - 2023

HTTP Version v1.1 v2.0 v3.0

Percentage



What are QUIC and HTTP/3?

Broadly speaking, QUIC is a **replacement** for the Transmission Control Protocol (TCP), one of the main protocols for internet communication. QUIC was originally developed internally by Google as Google QUIC, or gQUIC, and was presented to the IETF in 2015. Since then, it has been redesigned and improved by the broader IETF community, forming a new protocol we now call QUIC. **HTTP/3** is the next iteration of HTTP, the standard protocol for web-based applications and servers. Together, QUIC and HTTP/3 represent the latest and greatest in internet-focused protocols, incorporating decades of best practices and lessons that we, Google, and the IETF community learned through running protocols on the internet.

QUIC and HTTP/3 generally outperform TCP and HTTP/2, which in turn outperform TCP and HTTP/1.1. TCP and HTTP/2 first introduced the concept of allowing a single network connection to support multiple data streams in a process called stream multiplexing. QUIC and HTTP/3 take this one step further by allowing streams to be truly independent by avoiding TCP's dreaded head of line blocking, where lost packets jam and slow down all streams on a connection.

"experimental" phase and now carries a significant portion of global internet traffic. [↗](#)

As of early 2026, the usage statistics generally break down into **website support** (how many servers speak it) and **traffic volume** (how many actual bytes/requests use it).

1. Global Usage Statistics

- **Traffic Volume:** Approximately **32.5% to 33%** of global HTTP requests now travel over QUIC (HTTP/3).
- **Website Support:** About **37%** of all websites support HTTP/3 over QUIC. This number rises to roughly **44%** when looking at the top 1,000 most popular websites. [↗](#)

Note on Terminology: You will often see "HTTP/3" and "QUIC" used interchangeably in these stats. HTTP/3 is the application layer (the "carrier" of web pages) that *must* run on top of the QUIC transport layer. If a connection is HTTP/3, it is using QUIC.

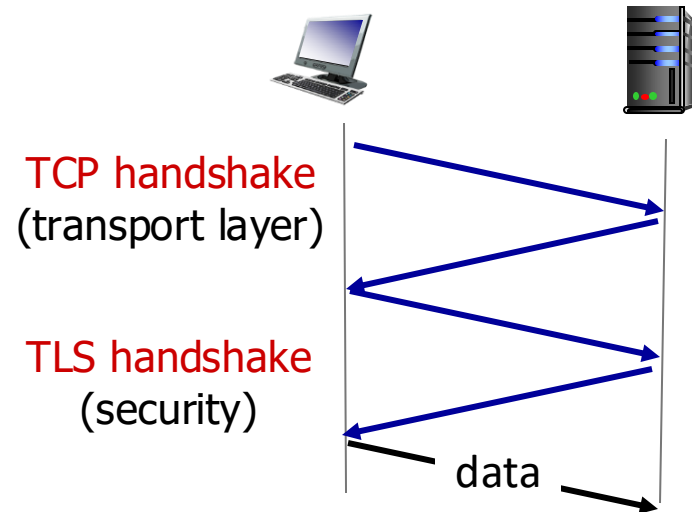


QUIC: Quick UDP Internet Connections

adopts approaches we've studied in this chapter for connection establishment, error control, congestion control

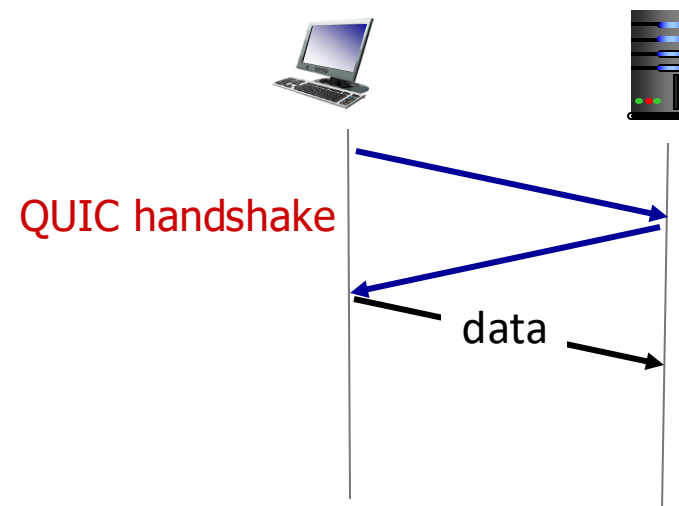
- **error and congestion control:** “Readers familiar with TCP’s loss detection and congestion control will find algorithms here that parallel well-known TCP ones.” [from QUIC specification]
- **connection establishment:** reliability, congestion control, authentication, encryption, state established in one RTT
- multiple application-level “streams” multiplexed over single QUIC connection
 - separate reliable data transfer, security
 - common congestion control

QUIC: Connection establishment



TCP (reliability, congestion control state) + TLS (authentication, crypto state)

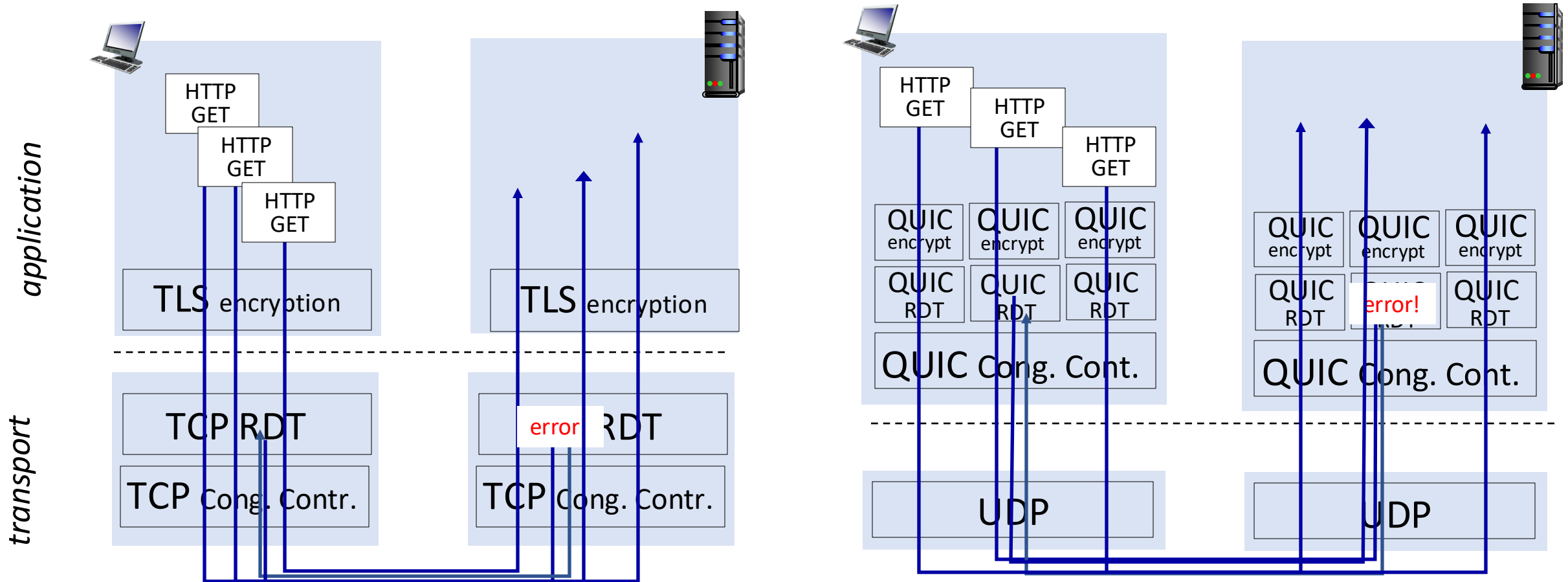
- 2 serial handshakes



QUIC: reliability, congestion control, authentication, crypto state

- 1 handshake

QUIC: streams: parallelism, no HOL blocking



(a) HTTP 1.1

(b) HTTP/2 with QUIC: no HOL blocking

Chapter 3: summary

- principles behind transport layer services:
 - multiplexing, demultiplexing
 - reliable data transfer
 - flow control
 - congestion control
- instantiation, implementation in the Internet
 - UDP
 - TCP

Up next:

- leaving the network “edge” (application, transport layers)
- into the network “core”
- two network-layer chapters:
 - data plane
 - control plane